Arunabha Majumder



Mechanical / Mechatronics Engineer

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PROFILE

Hands-on mechanical and mechatronics engineer who enjoys building things. I design, prototype and build robotic hardware, from concept and CAD (SolidWorks) through 3D printing, fabrication, mechanical assembly, testing and troubleshooting. I integrate sensors, actuators, motors and Arduino-based control electronics, and I built a Go-Kart and a hybrid vehicle in student competition teams.

EXPERIENCE

PhD Researcher, Robotics Hardware (project VIEXO)
Aalborg University, Denmark | *2023 - Present*

- Designed, prototyped and built a shoulder exoskeleton: custom mechanical components in SolidWorks, 3D printed, assembled and rebuilt over many iterations. It cut energy use by more than 25% (Best Paper Award, IFToMM ISRM 2026).
- Integrated a motor, encoder and force sensors with the mechanical frame, and measured the stiffness it provides on a physical test setup.
- Designed and modelled a novel 5-bar planar parallel robot with variable stiffness mechanisms; estimated its stiffness with less than 10% error and quantified human arm impedance.
- Implemented an Arduino-based trigger system to synchronise robot motion with Qualisys motion capture; collected EMG and motion data (data acquisition) from 12 participants.
- Troubleshooting and root cause: a clutch part kept breaking and was hard to make, so I redesigned it as a simple bolt-locking clutch that was easier to fabricate and reliable.
- Optimised the design with a genetic algorithm; taught courses and supervised Bachelor's students.

MTech Researcher, Pneumatic Artificial Muscles
CSIR-CMERI, India | *2020 - 2022*

- Fabricated and assembled pneumatic artificial muscles (pneumatics, valves, pressure sensors, DAQ) and collected test data on the lab's setup.
- Designed the outer PID loop of a neural network-based gain-scheduled controller in MATLAB/Simulink and implemented it in LabVIEW to drive the pneumatic muscle (0.3 - 0.78% error, IEEE CONECCT 2022).

Bachelor's Capstone and Student Teams
Siddaganga Institute of Technology, India | *2017 - 2019*

- Designed and built an upper-limb rehabilitation exoskeleton: CAD, machining, mechanical assembly and testing (Best Major Project Award, TATA Technologies grant).
- Student competition teams: built a Go-Kart (2017) and an Efficycle hybrid vehicle (2018) in the workshop with hand and power tools.

SKILLS

CAD and design: SolidWorks (CSWA), mechanical design, materials and tolerances, design for assembly

Prototyping and fabrication: 3D printing (Bambu Lab, Ultimaker Cura), basic machining, mechanical assembly, hand and power tools

Mechatronics: sensor integration, actuators, motors, encoders, force sensors, Arduino, control systems (PID), pneumatics, data acquisition, testing and troubleshooting

Software: MATLAB/Simulink, LabVIEW, Python (basic) | **Drawing:** hand sketching, technical illustration

EDUCATION

PhD, Mechatronics and Robotics, Aalborg University, Denmark | 2023 - present (research complete)

MTech (MSc), Mechatronics, AcSIR - CSIR-CMERI, India | 2020 - 2022

BTech (BSc), Mechanical Engineering, Siddaganga Institute of Technology, India | 2015 - 2019

Senior Diploma in Fine Arts (Painting), Bangiya Sangeet Parishad | 2008, Distinction

LANGUAGES AND REFERENCES

English (C1), Bengali (native), Hindi (spoken)

References: Prof. Shaoping Bai (sbh@mp.aau.dk); Assoc. Prof. Lars Rosgaard Jensen; Assoc. Prof. Anderson de Souza Castelo Oliveira; all Department of Materials and Production, Aalborg University